

Amazon SageMaker

Build, Train & Deploy ML Models, Fully Managed

AWS Machine Learning Associate • Part 11

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What We'll Cover



What is Amazon SageMaker?



Preparing & building



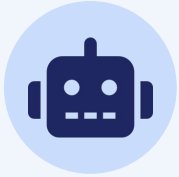
Training & tuning



Deploying models



Monitoring & automating



Shortcuts & key benefits

What is Amazon SageMaker?



Fully managed ML platform

- **Build, train & deploy ML models**
  Everything from data prep to production in one service
- **Fully managed**
  AWS runs the servers, scaling & infrastructure
- **Scales with your needs**
  From a small experiment to large distributed training

Naming note: renamed Amazon SageMaker AI in December 2024. You will see both names, and the ML features covered here are the same.

The ML Lifecycle in One Place

1



Prepare

Clean & label data

2



Build

Explore & write code

3



Train

Fit the model

4



Tune

Improve performance

5



Deploy

Serve predictions

6



Monitor

Watch for drift

SageMaker has tools for every one of these stages.

Prepare & Build



SageMaker Studio

A web-based environment for code, notebooks, and managing your ML work



Data Wrangler

Import, explore & transform data visually: missing values, encoding, scaling



Feature Store

A central place to save & reuse features across teams and models



Ground Truth

Label data with human reviewers and automation

Everything from the last few videos, in managed tools.

Train & Tune



Managed Training Jobs

Point to data in S3; SageMaker runs the compute, saves the model, shuts down



Your Choice of Code

Built-in algorithms like XGBoost, frameworks like PyTorch, or your own container



Automatic Model Tuning

Runs many training jobs to find the best hyperparameters for you



Managed Spot Training

Use spare compute capacity to cut training costs significantly

Data in, model out, machines off.

Deploy: Inference Options



Real-Time Endpoints

Always-on, low-latency predictions. Best for: fraud checks at purchase



Serverless Inference

Scales automatically, even to zero. Best for: bursty traffic



Asynchronous Inference

Queues requests. Best for: large payloads & long processing times



Batch Transform

Predictions over a whole dataset, no endpoint to keep running

Choose by latency needs, traffic pattern, and cost.

Monitor & Automate



Model Monitor

Watches live predictions & alerts you to data drift



Clarify

Detects bias and explains why a model made a prediction



Model Registry

Tracks model versions & approvals before production

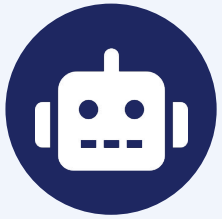


Pipelines

Automates the whole workflow as a repeatable process

Automating this lifecycle is what people call MLOps.

Shortcuts for a Faster Start



Autopilot

AutoML. Give it a dataset and a target; it tries algorithms & settings and shows its work.



JumpStart

A hub of pretrained & foundation models and ready-made solutions to deploy in a few clicks.



Canvas

A no-code, point-and-click interface so business analysts can build predictions.

Key Benefits



Fully Managed

No servers to maintain



Scales Easily

From a small experiment to large distributed training



Pay for What You Use

Training machines shut down when jobs finish



Integrated with AWS

S3 for data, IAM for permissions, CloudWatch for logs & metrics

And it covers the whole lifecycle, so no stitching tools together.

Recap

- SageMaker is AWS's fully managed platform for the whole ML lifecycle.
- Prepare, train, tune, deploy, and monitor, all in one place.
- Autopilot, JumpStart & Canvas give you shortcuts along the way.

Thanks for watching!